MOHAMMAD AMIR KHAN

Software Engineer | C++ / Qt Developer

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

C++/Qt Software Engineer with 1+ year of experience building 2 enterprise endpoint-security and remote-access products for Windows and macOS. Skilled in C++11/14/17, Qt 5, WinAPI, multithreading, TCP/IP, REST APIs and SQLite. Delivered DLP, employee monitoring, live screen casting and secure remote desktop features, with up to 97% lower latency and GUI freezes eliminated.

TECHNICAL SKILLS

Languages: C++ (C++11/14/17), C, Objective-C++, Python, Java, JavaScript, SQL, Shell Scripting

Qt & Desktop: Qt 5.15, Qt Widgets, QML, Qt Network, Qt WebSockets, Qt SQL, QtConcurrent, QThread, Signals & Slots, WinAPI, Cocoa/AppKit

Windows & Systems: Windows Hooks, UI Automation, Windows Services, Shared Memory, Named Pipes, IPC, Multithreading, Performance Optimization

Networking & Security: TCP/IP, WebSocket, REST APIs, JSON, HTTPS/TLS Proxy, SMTP, IMAP, VNC/RFB, SHA-256, Data Loss Prevention (DLP)

Core CS Design: Data Structures & Algorithms, Operating Systems, Computer Networks, OOP, Design Patterns, UML, Requirement Analysis

Databases & AI: SQLite, SQL, MongoDB

Testing & Tools: Unit Testing, CppUTest, Integration Testing, Git, GitHub, CMake, Qt Creator, Visual Studio, Agile/Scrum

Platforms: Windows, macOS, Linux

PROFESSIONAL EXPERIENCE

Software Developer | Globussoft Technology Pvt. Ltd. *Aug 2025 – Present*

EmpMonitor – Enterprise Employee Monitoring & DLP Platform (Windows)

Tech Stack: *C++, Qt 5, WinAPI, UI Automation, Windows Hooks, SQLite, HTTPS/TLS Proxy, WebSocket, REST*

- Built a Windows monitoring agent in C++/Qt covering email, clipboard, keystrokes, Bluetooth, screenshots and screen recording.
- Developed email monitoring through an HTTPS/SMTP/IMAP proxy for Gmail, Zoho, Thunderbird and Outlook.
- Cut duplicate email dashboard rows by ~67% (3 to 1 per send) using subject canonicalization and a 15 s dedup window.
- Removed a 1.2 s per-connection stall in the email proxy by moving TLS handshakes to a thread pool.
- Implemented file-upload DLP using UI Automation, a WH_CALLWNDPROC hook DLL and shared-memory policy sync.
- Blocked drag-and-drop uploads and stored detections offline-first in SQLite with batched reporting (500-entry queue, 3 s).
- Developed app and website blocking with exact-host matching and a 5 s debounce to prevent false tab closures.
- Cut UI Automation polling by up to ~90% and file-upload worker ticks by 80% through tiered, longer timers.
- Eliminated 0.5–2 s GUI freezes per screenshot by moving capture off the GUI thread.
- Reduced clipboard dashboard latency by up to 97% (180 s to 5 s) with a 5 s single-shot flush.
- Built live screen casting with a LocalSystem service for UAC and lock-screen capture.
- Developed 2 webcam-based detection workflows: user-presence and device/phone detection.
- Cut remote-input round trips by ~58% by coalescing mouse moves (60 Hz to ~25 Hz) and sending click/drag in one round trip.
- Integrated REST APIs with JSON-based communication between the desktop agent and backend services.
- Added a 450 MB memory watchdog that automatically relaunches the agent.

Secure RDP Controller – Secure Remote VM Access (macOS Host App + Agent Daemon)

Tech Stack: *C++17, Qt 5, Objective-C++, Cocoa/AppKit, libvncclient, QWebSocket, TCP/IP, JSON*

- Built a Qt/Objective-C++ host app for secure remote VM access with VNC/RFB display on a dedicated thread.
- Developed a C++ agent daemon with UTM VM lifecycle control and a 2 s IP-readiness poll.
- Designed a newline-delimited TCP/JSON control protocol (port 5890) that handles partial and coalesced reads.
- Implemented secure tunneled communication for remote VM connectivity and Host-Agent interaction.
- Implemented TCP/IP socket communication and JSON-based messaging between Host and Agent applications.
- Implemented chunked file/folder transfer with SHA-256 integrity checks and 1 MiB streamed reads.
- Enforced 4 directional clipboard policies (text and file, VM-to-host and host-to-VM).
- Built a WebSocket gateway proxy with AUTH_INIT, bearer token, heartbeat and retry.
- Implemented session inactivity timeout with a 20 s grace period and a 2 s activity filter.
- Implemented screen-capture protection, watermarking and 2 s foreground-app monitoring using Cocoa and Accessibility APIs.
- Built script scheduling with host approval and OS filtering (zsh, bash, cmd.exe) using QProcess.

EDUCATION

Bachelor of Engineering, Computer Science Engineering | Visvesvaraya Technological University (CGPA: 8.0) *2021 – 2025*